Class Meeting Handout #6

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Our review did not end in meeting #5. We will spend some time today reviewing acid/base equilibria as it applies to organic chemistry mechanism.

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Acid-Base Equilibrium

We will then explore how to measure the acid "power" of a mixture with scales that go beyond *p*H.

Review: Brønsted-Lowry Theory – 20 minutes

You will have discussed Brønsted-Lowry acid theory in almost every chemistry class that you have had so far. Let us have fun with the following class problems as we review.[1]

1. Compare the expressions for K_{eq} and K_a of acid dissociation in water.
2. Write our the chemical equation for the acid equilibrium of acetic acid. Write out the equation for the acid dissociation equilibrium constant, K_a.
3. Write our the famous Henderson-Hasselbalch equation and show me how it is just an alternate version of the acid equilibrium expression.
4. The pK_a value for acetic acid is 4.75. What is the value of the equilibrium constant for acid dissociation, K_a
5. Tell me how you would mix up a 100 mM acetate buffer so that when you measure the *p*H it will be at or very near a *p*H value of 5.1.

[1] Chapters 5.1 and 5.2 of the textbook are useful reading as you review.

Table 1: Selected pK_a values.

Acid	pK_a
Acetic acid	4.75
Phenol	9.95
4-Nitrophenol	7.10
Saccharine	1.60
Aniline	4.62
4-aminoazobenzene	2.82
4-nitroaniline	1.11
4-Chloro-2-nitroaniline	−1.02

Discussion: Extreme Acid-Base Equilibrium – 10 minutes

*p*H can only get us to 1% acid. What if we want to explore a mixture that is 99% acid? The Hammett acidity function, H_0, can describe acidity far beyond the range of *p*H.[2,3,4]

6. What is the protonating power as expressed in *p*H units of 1 L of water with enough glacial acetic acid added to make a solution with a total concentration of 0.01 M.
7. How would you determine the protonating power of pure glacial acetic acid? Would a *p*H electrode be accurate here?
8. 4-chloro-2-nitroaniline is 50% protonated in pure glacial acetic acid. Express the protonating power of pure acetic acid using an acidity function. Describing the activity of acid mixtures is important.[5,6]

[2] Acidity functions are presented in chapter 5.2.5 of the textbook.

[3] The Curtin-Hammett principle, the Hammett acidity function, the Hammett equation, the Hammett substituent parameters – is there anything that Louis Hammett didn't touch in the field of Physical organic chemistry? I suppose thats what happens when you are the founding father. Below is the equation for the Hammett acidity function, H_0

$$H_0 = pK_a + \log\frac{[A]}{[AH]}$$

[4] IUPAC limits the *p*H acidity function to between $2 \le pH \le 12$.

[5] From before even I was born: "The Acidity Scale in Glacial Acetic Acid. I. Sulfuric Acid Solutions. $-6 < H_0 < 0$." N.F. Hall, W.F. Spengeman *J. Am. Chem. Soc.*, **1940**, *62*, 2487–2492 https://doi.org/10.1021/ja01866a062

[6] Compare the above work to this, eighty years later: "The determination of the Hammett acidity function H_0 in a mixtures of phosphoric and acetic acids by NMR measurements." T.Y. Ivanenko, A.A. Kondrasenko, A.I. Rubaylo, *Journal of Molecular Liquids*, **2023**, *391B*, 123438. https://doi.org/10.1016/j.molliq.2023.123438

Exercise: Going to the Extreme – 10 minutes

We will only be using the Hammett acidity function, H_0, for describing the acidity of strong acid mixtures. There are many more. The choice of a given acidity function will depend on the situation. H_0 values are determined using aniline dyes and are suitable for polar second row atoms (oxygen, nitrogen) in high dielectric conditions.[7] Be prepared to be criticized for your choice.[8]

[7] For a useful review see "Acidity functions: an update." R.A. Cox, K. Yates, *Can. J. Chem.*, **1983**, *61*, 2225–2243. https://doi.org/10.1139/v83-388

[8] For more examples of acidity function scales, see "Protonation and Solvation in Strong Aqueous Acids." E.M. Arnett, G. Scorranno, *Adv. Phys. Org. Chem.*, **1976**, *13*, 83–153. https://doi.org/10.1016/S0065-3160(08)60213-0. Accessed via inter-library loan. Ask me for a copy.

9. What is the *p*H of 0.1% (w/w) HNO_3 in water? What is the *p*H of 85% (w/w) HNO_3 in water?[9] Use the data below to answer this question.[10] Compare your results to the chart in figure 1.

[9] The molar mass of 100% HNO_3 is 63.01 g/mole and the density is 1.51 g/mL

	% in acidic form			
% w/w HNO_3	4-nitro-aniline	4-chloro-2-nitroaniline	4,6-dichloro-2-nitroaniline	6-bromo-2,4-dinitroaniline
0.1	14.53	0.15	0.00	0.00
4	92.80	10.30	0.03	0.00
40	99.96	95.63	5.32	0.01
85	100.00	99.99	95.72	2.05
100	100.00	100.00	99.80	31.37

Table 2: Observed ionization of selected indicator acids in aqueous mixtures of HNO_3. See table 1 for useful pK_a values

[10] Values in table 2 calculated from data within "The Hammett acidity function in aqueous nitric acid." J.G. Dawber, P.A.H. Wyatt, *J. Chem. Soc.*, **1960**, 3589–3593. https://doi.org/10.1039/JR9600003589. Available in UPEI library bound periodicals.

10. As revealed in figure 1, the acidity functions of $HClO_4$, HCl, HNO_3 & H_2SO_4 are identical in dilute aqueous solution. Why?

Learning Goals

There is more to a class meeting than what we do in 50 minutes of class time. After participating in the above exercises and completing the requirements described on the moodle site you will have achieved the following...

1. Reviewed Brønsted-Lowry acid-base theory.
2. Be able to use the Henderson-Hasselbalch equation and the Hammett acidity function to describe the protonating power of acidic systems.
3. Understand the caveats and limits of the Hammett acidity function and similar acidity functions.

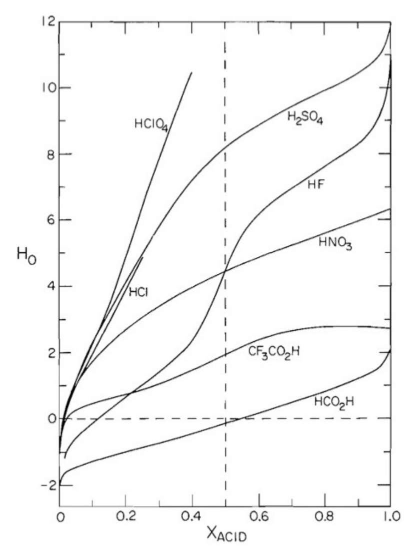

Figure 1: H_0 values vs. mole fraction for selected mixtures of acids and water.[7]

Looking Ahead

If we want to understand the kinetics of reactions that occur in acid (or base) then we need to be able to accurately describe the activity (protonating power) of the acid. If we double the activity, we would double the rate if the acid is part of the rate law. If you want to plot rate vs. acid activity you need an accurate scale. *p*H works well in dilute aqueous solutions but when we are following reactions in 50% H_2SO_4/40% ACN/10% H_2O we need an acidity function. Choose wisely (we will always choose H_0.)